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Max–min fairness driven multicast sparse beamforming for cache-enabled Cloud RAN

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ABSTRACT

Cloud radio access network (C-RAN) is able to boost network capacity through centralized scheduling of multiple distributed base stations (BSs). Yet it also places serious burdens on backhaul links. Caching and multicast are two enabling candidates to alleviate backhaul cost and achieve efficient content delivery. In this paper, we consider a cache-enabled multicast C-RAN, where multiple BSs cooperatively serve multiple user groups. Each BS is equipped with local storage and is connected to the central processor (CP) via a backhaul link. Then, the beamforming vectors and dynamic BS clustering is carefully designed to maximize fairness among users. To achieve such a goal, a semidefinite relaxation (SDR)-based iterative difference-of-two-convex-function (D.C.) algorithm is proposed by using semidefinite programming (SDP) and smoothed l_0-norm approximation approaches. After that, to reduce computational complexity, a two-tier convex quadratic-based alternating (TCQA) algorithm is devised by decoupling multi-ratio fractional constraints. In this algorithm, the outer problem is handled in closed-form expressions while the inner one is solved by using D.C. programming. Convergence and complexity of the two proposed algorithms are analyzed. Finally, extensive simulation results demonstrate that our proposed TCQA algorithm significantly outperforms the SDR-based iterative D.C. algorithm.

1. Introduction

The exponential growth of mobile devices has put forward higher requirements for the base station (BS) to support higher data traffic and connect more users. Recently, researchers have devised various transmission schemes and network architectures to shorten latency and stimulate the system throughout. One of the promising architectures is cloud radio access network (C-RAN) [1–3]. However, the bottleneck of C-RAN is that backhaul links are resource-limited. Meanwhile, traffic congestion may still occur during the peak time [4]. Caching, as a promising solution to further reduce transmission delay and alleviate backhaul links burden, has attracted considerable attention from both academia and industry [5–9].

Cache-enabled wireless networks can pre-fetch required contents through distributed storages during the off-peak time. The content is transmitted when subscribers submit requests, so traffic congestion can be greatly alleviated. At the same time, in practical application scenarios, data traffic is mainly controlled by a few popular contents, which are requested by many mobile users. To transmit such contents more effectively, an enabling technology is multicast. Specifically, compared with conventional point-to-point unicast communication, multicast can transmit common content to multiuser without inter-user interference power. More performance analyses and optimizations involving cache-aided multicast technique are presented in [10–13].

These previous works presented content-centric cache-enabled multicast transmission strategies without considering beamformers' optimization. Recently, cache-enabled content delivery schemes by carefully designing multicast beamforming vectors and dynamic BS clustering have been discussed. The several major criteria for evaluating cache-enabled multicast network are minimizing latency [14,15], the optimizing weighted total network costs [16–18], and maximizing the system throughout or energy efficiency [19–22]. In [14], the total latency for backhaul and access link backhaul is optimized for a two-tier ultra-dense wireless network. In [15], a content delivery strategy considering min–max delay is derived by jointly optimizing the fronthaul bandwidth allocation, multicast beamforming, and BS clustering. In [17], a content-centric transmission strategy is designed to minimize the weighted sum of transmission power and backhaul cost for a cache-enabled multicast C-RAN. In [18], multicast mode, caching strategy, and sparse beamforming vectors are jointly optimized for minimizing the network power cost subject to users' quality of service (QoS). A

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similar network model is considered in [19], but its emphasis is to optimize energy efficiency by incorporating the beamformers and BS clustering. In [23], the multicast beamforming vectors as well as cache size allocation strategy are derived respectively in two timescales for a downlink C-RAN. The research efforts mentioned above are mainly to reduce the backhaul links load or shorten the transmission delay, but the fairness among users' achievable rate, as an important indicator, is rarely considered except [24]. In [24], the worst-user's rate among multicast groups is maximized by solving a power minimization problem and using the bisection method. However, the proposed algorithm has high complexity and is not suitable for each BS's power constraint. This is because the formulated problem subject to each BS's power constraint cannot be recast to a similar power minimization problem formulated in [24]. Therefore, such a situation drives us to study the max–min fairness among all users for a cache-enabled multicast C-RAN.

In this paper, with the goal of maximizing the worst-user's rate for a C-RAN with limited transmission power and backhaul cost, we jointly design dynamic BS clustering and multicast beamforming vectors. However, the formulated problem is a nonlinear programming problem, which is mathematically challenging to attain optimal sparse precoders with analytic expressions. To exploit the convexity, two algorithms are devised, namely, semidefinite relaxation (SDR)-based iterative difference-of-two-convex-function (D.C.) algorithm and two-tier convex quadratic-based alternating (TCQA) algorithm. Compared to another max–min fairness problems for a C-RAN [24], the main contributions of this work are summarized as follows,

- We establish a cache-enabled downlink multicast C-RAN in the presence of one macro BS (MBS) and multiple small BSs (SBSs) with the limited cache storage and coverage area. After that, the joint beamforming vectors design and BS scheduling problem is formulated with each BS's power constraint. The devised algorithm in [24] assuming total power constraint cannot be used to our optimization problem, which further motivates the algorithmic investigations in this paper.
- To overcome the non-convexity caused by beamforming vectors coupling and binary integer variables, semidefinite programming (SDP) and successive convex approximation (SCA) are employed to recast the problem into a convex form. Meanwhile, a re-weighted l_1-norm is used to solve binary BS scheduling. Then, the transformed problem can be effectively solved by the proposed local optimal iterative D.C. algorithm.
- In order to reduce computational complexity, we develop the multi-ratio quadratic transform method to reconstruct the formulated problem into a two-tier alternating problem. After that, the outer-tier is solved in closed-form expressions while the inner-tier is locally optimized by using the D.C. algorithm. Furthermore, the convergence of the proposed TCQA algorithm is proved.

The remainder of the paper is organized as follows: In Section 2, we elaborate the system model, the cache model, and problem formulation. After that, the two joint beamformers and dynamic BS clustering optimization algorithms are developed and analyzed in Section 3. Furthermore, the convergence and complexity of the SDR-based iterative D.C. algorithm as well as the TCQA algorithm are computed. In Section 4, simulation results are shown to validate the effectiveness of the proposed algorithms. Finally, a summary is provided in Section 5.

Notations. We use uppercase and lowercase boldface letters to denote matrices and vectors, respectively. $(\mathbf{A})^H$, $\mathrm{Tr}(\mathbf{A})$ and $\mathrm{Rank}(\mathbf{A})$ are Hermitian transpose, trace and rank of matrix $\mathbf{A}$, respectively. Furthermore, $\mathbf{A} \succeq \mathbf{0}$ denotes that $\mathbf{A}$ is a positive semidefinite matrix while $\mathcal{CN}(\mu, \sigma^2)$ stands for the μ-mean σ^2-variance complex Gaussian distribution. Besides, b^* stands for the optimum of variable b. Finally, $|\mathcal{S}|$, $\left\|\mathbf{b}\right\|_p$ and $\mathrm{Re}(\mathbf{b})$ stand for the cardinality of $\mathcal{S}$ and l_p-norm and the real part of vector $\mathbf{b}$, respectively.

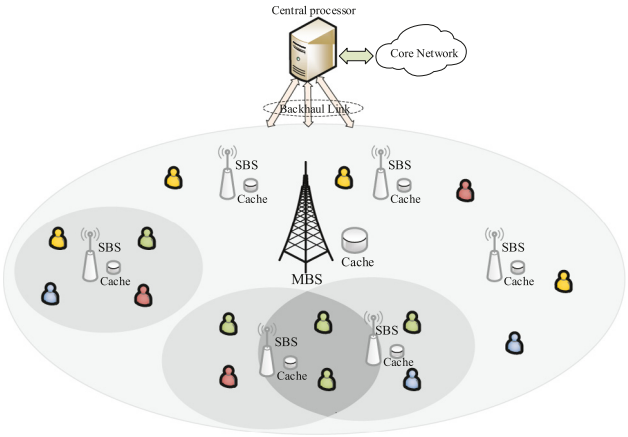


Fig. 1. Cached-enabled downlink multicast C-RAN with capacity-limited backhaul links.

2. System model and problem formulation

2.1. System model

In this paper, we consider the cache-enabled downlink multicast C-RAN in the presence of one MBS and K SBSs collaboratively transmitting contents to L single-antenna users, as shown in Fig. 1. The MBS and each SBS are equipped with M and N antennas, respectively. We assume that MBS can pre-fetch all contents and serve all users, but each SBS with limited storage space can only cache part of contents. Each SBS is connected to the central processor (CP) via a capacity-limited backhual link. In this work, we consider that the coverage area of SBS is limited. Let $\mathcal{K} = \{0, 1, 2, \ldots, K\}$ and $\mathcal{L} = \{1, 2, \ldots, L\}$ denote the BS and user index set, respectively, where $k = 0$ denotes the MBS. At the beginning of scheduling interval, users submit content requirements. Users requesting the same content are collected to a multicast group and served by a cluster of BSs according to its requirements and locations. Let $\mathcal{S} = \{1, 2, \ldots, S\}$ denote the set of multicast groups and $S = |\mathcal{S}|$ is the cardinality of $\mathcal{S}$. We assume that each user can only request at most one content in a scheduling slot, which means that $\sum_{s=1}^{S} |\mathcal{L}_s| \le L$ and $\mathcal{L}_s \cap \mathcal{L}_{s'} = \emptyset$ for $\forall s \ne s'$, where $\mathcal{L}_s$ denotes the user set included in the sth multicast group.

We define a binary matrix $\mathbf{D} \in \{0,1\}^{K \times L}$, where $d_{k,l} = 1$ if there exists a communication link between the kth SBS and the lth user, otherwise $d_{k,l} = 0$, where $k \ne 0$. Let $\mathbf{h}_l = \left[\mathbf{h}_{0,l}^H, d_{1,l}\mathbf{h}_{1,l}^H, \ldots, d_{K,l}\mathbf{h}_{K,l}^H\right]^H \in \mathbb{C}^{(M+NK)\times 1}$ denote the channel vector from all BSs to the lth user, where $\mathbf{h}_{k,l} \in \mathbb{C}^{M\times 1}$ or $\mathbb{C}^{N\times 1}$ denotes the channel vector from BS k to the lth user for $\forall l \in \mathcal{L}$. Similarly, the beamforming vector of the sth multicast group is denoted by $\mathbf{w}_s = \left[\mathbf{w}_{0,s}^H, \mathbf{w}_{1,s}^H, \ldots, \mathbf{w}_{K,s}^H\right]^H \in \mathbb{C}^{(M+NK)\times 1}$, where $\mathbf{w}_{k,s} \in \mathbb{C}^{M\times 1}$ or $\mathbb{C}^{N\times 1}$ is the beamforming vector of BS k in the sth multicast group. If BS k is not selected to serve the sth multicast group, precoding vector $\mathbf{w}_{k,s}$ should be set to $\mathbf{0}$ for minimizing backhaul cost.

Therefore, the signal at the lth receiver in the sth multicast group can be modeled as

$$y_l = \mathbf{h}_l^H \mathbf{w}_s x_s + \sum_{j=1, j\ne s}^{S} \mathbf{h}_l^H \mathbf{w}_j x_j + n_l, \tag{1}$$

for $\forall l \in \mathcal{L}_s$ and $\forall s \in \mathcal{S}$, where $n_l \sim \mathcal{CN}(0, \sigma_l^2)$ is additive white Gaussian noise (AWGN) with zero-mean and σ_l^2-variance. The transmitted signal has been normalized, *i.e.* $\mathbb{E}|x_s|^2 = 1$ for $\forall s \in \mathcal{S}$. Then, the resulting signal to interference and noise ratio (SINR) can be calculated as

$$\gamma_l = \frac{\left|\mathbf{h}_l^H \mathbf{w}_s\right|^2}{\sum_{j=1, j\ne s}^{S} \left|\mathbf{h}_l^H \mathbf{w}_j\right|^2 + \sigma_l^2}, \quad \forall l \in \mathcal{L}_s, \forall s \in \mathcal{S}. \tag{2}$$

2.2. Cache model and problem formulation

We first define a binary cache matrix $\mathbf{C} \in \mathbb{C}^{K\times F}$, $c_{k,f} = 1$ if BS k has cached the fth content, otherwise $c_{k,f} = 0$, where $k \neq 0$. In general, cache placement needs larger timescale than users/BSs scheduling and transmission. Thus, we consider that the cache placement content is known and fixed. Then, we focus on beamforming and dynamic BS clustering design to maximize the worst-user's rate. Similarly, we introduce a binary BS selection matrix $\mathbf{Q} \in \mathbb{C}^{K\times S}$, $q_{k,s} = 1$ indicates BS k is chosen to cooperatively serve the sth multicast group, otherwise $q_{k,s} = 0$ for $k \neq 0$.

The content required by the sth multicast group is denoted by f_s. Therefore, if $q_{k,s} = 1$ and $c_{k,f_s} = 0$ are true at the same time, BS k requests content f_s from CP via a backhual link, which results in backhaul cost. We model the total backhaul cost as the accumulated amount of unique data transferred from the CP, which can be modeled as:

$$C_k = \sum_{s=1}^{S} \left(1 - c_{k,f_s}\right) q_{k,s} \log(1 + \Gamma_s). \tag{3}$$

Since the sth multicast group rate is determined by the worst-user in $\mathcal{L}_s$, $\Gamma_s = \min_{l\in\mathcal{L}_s} \gamma_l$ is valid for alleviating backhaul cost.

Finally, we formulate the max–min fairness problem among all users by jointly designing beamforming vectors and dynamic BS clustering subject to each BS power and backhaul cost. Thus, the worst-user's rate maximization problem can be written as

$$\max_{\mathbf{w}_{k,s}, q_{k,s}} \min_{s\in\mathcal{S}} \min_{l\in\mathcal{L}_s} \gamma_l \tag{4a}$$

subject to

$$\sum_{s=1}^{S} \|\mathbf{w}_{k,s}\|^2 \leq P_k, \quad \forall k \in \mathcal{K}, \tag{4b}$$

$$\sum_{k=1}^{K} C_k \leq C_{th}, \tag{4c}$$

$$(1 - q_{k,s})\mathbf{w}_{k,s} = 0, \quad \forall k \in \mathcal{K}, \forall s \in \mathcal{S}, \tag{4d}$$

$$q_{k,s} \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \forall s \in \mathcal{S}, \tag{4e}$$

where (4b) is the power constraint for each BS while (4c) is the total backhual cost constraint. Furthermore, (4d) is imposed for reducing backhaul links load, and (4e) indicates the discrete nature of optimization variables. Finally, constant P_k in (4b) and C_{th} in (4c) denote the resource threshold.

3. Proposed algorithms

In this section, we mainly design two iterative algorithms to tackle the max–min fairness problem. The proposed algorithms are named SDR-based iterative D.C. algorithm and two-tier convex quadratic-based alternating algorithm, respectively. Furthermore, we analyze and prove some properties of the proposed algorithms, such as convergence, complexity, and optimality.

3.1. SDR-based iterative D.C. algorithm

Here, we design the SDR-based iterative D.C. algorithm to jointly optimize the beamforming vectors and dynamic BS clustering. Specifically, re-weighted l_1-norm and SCA methods are applied to handle binary BS scheduling and non-convex backhaul cost constraint. In this algorithm, we avoid matrix derivation calculation by introducing variables.

To proceed, we first introduce a set of matrices $\mathbf{W}_s = \mathbf{w}_s\mathbf{w}_s^H \succeq \mathbf{0}$ for $\forall s \in \mathcal{S}$. Let $\mathbf{H}_l = \mathbf{h}_l\mathbf{h}_l^H$ denote the new channel matrices for $\forall l \in \mathcal{L}$. We then remove the minimum operators by introducing auxiliary variable t, so (4) can be recast as

$$\max_{\mathbf{W}_{k,s}, q_{k,s}, t} \quad t \tag{5a}$$

subject to

$$\frac{\mathrm{Tr}(\mathbf{H}_l\mathbf{W}_s)}{\sum_{j=1,j\neq s}^{S} \mathrm{Tr}(\mathbf{H}_l\mathbf{W}_j) + \sigma_l^2} \geq t, \quad \forall l \in \mathcal{L}_s, \forall s \in \mathcal{S}, \tag{5b}$$

$$\sum_{s=1}^{S} \mathrm{Tr}\left(\mathbf{W}_{k,s}\right) \leq P_k, \quad \forall k \in \mathcal{K}, \tag{5c}$$

$$(1 - q_{k,s})\mathrm{Tr}\left(\mathbf{W}_{k,s}\right) = 0, \quad \forall k \in \mathcal{K}, \forall s \in \mathcal{S}, \tag{5d}$$

(4c), (4e), (5e)

$$\mathrm{Rank}\left(\mathbf{W}_{k,s}\right) = 1, \quad \forall k \in \mathcal{K}, \forall s \in \mathcal{S}, \tag{5f}$$

$$\mathbf{W}_s \succeq \mathbf{0}, \quad \forall s \in \mathcal{S}, \tag{5g}$$

where (5f) is imposed because $\mathbf{W}_{k,s}$ must be decomposed into a one-dimensional vector.

The fractional inequality in (5b) involves a non-convex set. To overcome the challenge, we first make the following equivalent transformation for $\forall l \in \mathcal{L}_s$ and $\forall s \in \mathcal{S}$.

$$\mathrm{Tr}(\mathbf{H}_l\mathbf{W}_s) \geq t\left(\sum_{j=1,j\neq s}^{S} \mathrm{Tr}(\mathbf{H}_l\mathbf{W}_j) + \sigma_l^2\right). \tag{6}$$

However, (6) still cannot define a convex set because the mutual coupling between t and beamforming matrices [25]. Then, the new auxiliary variables $\mu = [\mu_1, \ldots, \mu_L]$ are introduced to decompose (6) into two equivalent constraints as follows.

$$\mathrm{Tr}(\mathbf{H}_l\mathbf{W}_s) \geq \mu_l^2, \tag{7a}$$

$$\frac{\mu_l^2}{t} \geq \sum_{j=1,j\neq s}^{S} \mathrm{Tr}(\mathbf{H}_l\mathbf{W}_j) + \sigma_l^2. \tag{7b}$$

Note that the inequality that determined by convex function is bigger than or equal to affine function remains non-convex set. To tackle this challenge, we adopt an iterative affine function to replace $\frac{\mu_l^2}{t}$. This affine function is its first-order Taylor expansion at the fixed point $\left(\mu_l^{(i)}, t^{(i)}\right)$, where $\left(\mu_l^{(i)}, t^{(i)}\right)$ is the initial point at the ith iteration and (i) is iteration index for $\forall l \in \mathcal{L}$.

$$\frac{2\mu_l^{(i)}\mu_l}{t^{(i)}} - \frac{{\mu_l^{(i)}}^2 t}{{t^{(i)}}^2} \geq \sum_{j=1,j\neq s}^{S} \mathrm{Tr}(\mathbf{H}_l\mathbf{W}_j) + \sigma_l^2, \tag{8}$$

The transformation from (7) to (8) is effective because the first-order Taylor expansion of convex function is less than or equal to itself. The credible domain is narrowed down, which means that the solution we get is a local optimum of the original problem. However, with the increase of iteration, the first-order Taylor expansion can gradually approach the original function because its second-order partial derivative is approximately equal to 0. Now, the constraint (5b) has been transformed into a convex set.

In order to represent $\mathbf{w}_{k,s}$ with $\mathbf{w}_s$, we define a set of binary diagonal selection matrices $\mathbf{J}_k$ for $\forall k \in \mathcal{K}$, which is expressed as

$$\mathbf{J}_k = \mathrm{diag}\left(\left[\underbrace{0, \ldots, 0}_{M+N(k-1)}, \underbrace{1, \ldots, 1}_{N}, \underbrace{0, \ldots, 0}_{N(K-k)}\right]\right). \tag{9}$$

We thus rewrite $\mathrm{Tr}\left(\mathbf{W}_{k,s}\right)$ as

$$\mathrm{Tr}\left(\mathbf{W}_{k,s}\right) = \mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right). \tag{10}$$

Furthermore, we can observe that the BS selection variable $q_{k,s}$ can be modeled as the knowledge of $\mathrm{Tr}\left(\mathbf{W}_{k,s}\right)$. Specifically, (5d) can be expressed as

$$q_{k,s} = \begin{cases} 1, & \text{if } \mathrm{Tr}\left(\mathbf{W}_{k,s}\right) \neq \mathbf{0}; \\ 0 \quad \text{or} \quad 1, & \text{if } \mathrm{Tr}\left(\mathbf{W}_{k,s}\right) = \mathbf{0}. \end{cases} \tag{11}$$

Summarizing (4c), (10) and (11), $q_{k,s} = 0$ must be established for reducing backhual cost when $\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right) = \mathbf{0}$. Therefore, according to

(11), the binary variable $q_{k,s}$ can be replaced with l_0-norm of $\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right)$

$$q_{k,s} = \left\|\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right)\right\|_0. \tag{12}$$

However, the discrete nature of the variable $q_{k,s}$ has not changed. To cope with this challenge, the non-convex discrete l_0-norm is approximately transformed into its re-weighted sparsity-promoting l_1-norm. Specifically, $\left\|\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right)\right\|_0 = \theta_{s,k}\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right)$, where $\theta_{s,k}$ is a constant weight associated with the last beamformers $\mathbf{W}_s^{(i)}$, which is updated iteratively as follows:

$$\theta_{s,k} = \frac{1}{\mathrm{Tr}\left(\mathbf{W}_s^{(i)}\mathbf{J}_k\right) + \delta} \tag{13}$$

where δ is a small constant for guaranteeing the meaningfulness of function.

Next, for alleviating the backhaul load and ensuring the fairness among users, the backhaul cost can be reduced to $C_k = \sum_{s=1}^{S}\left(1 - c_{k,f_s}\right) q_{k,s}\log(1+t)$ because the maximum of the worst-user's rate is not weakened. Therefore, the resulting backhaul cost constraint can be modeled as

$$\sum_{s=1}^{S}\left(1 - c_{k,f_s}\right)\theta_{s,k}\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right) \le \frac{C_{th}}{\log(1+t)}. \tag{14}$$

Meanwhile, since the function in right side of (14) is convex, we develop an approximate function by using first-order Taylor expansion around a fixed point $t^{(i)}$ and have

$$\begin{aligned}\sum_{s=1}^{S}\left(1 - c_{k,f_s}\right)\theta_{s,k}\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right) \le C_{th}\Bigg(&\frac{1}{\log(1+t^{(i)})}\\ &-\frac{1}{\log^2(1+t^{(i)})(1+t^{(i)})\ln 2}(t - t^{(i)})\Bigg),\end{aligned} \tag{15}$$

which can define a convex set.

By removing the rank-one constraint $\mathrm{Rank}\left(\mathbf{W}_{k,s}\right) = 1$ for $\forall k \in \mathcal{K}$ and $\forall s \in \mathcal{S}$, problem (5) can be recast as

$$\max_{\mathbf{W}_s, t} \quad t \tag{16a}$$

subject to

$$\sum_{s=1}^{S}\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right) \le P_k, \quad \forall k \in \mathcal{K}, \tag{16b}$$

$$(7a),\ (8),\ (15),\ (5g). \tag{16c}$$

At present, the recast problem (16) has been transformed into a convex problem, which can be solved by using any standard solvers, such as CVX [26]. However, owing to the rank-one constraint has been relaxed, the equivalence between the problem (4) and (16) cannot be ensured. Therefore, if the obtained sub-optimal $\mathbf{W}_{k,s}$ is already rank-one, the beamforming $\mathbf{w}_{k,s}$ can be calculated by eigen-value decomposition (EVD). Otherwise, the Gaussian randomization method is employed to generate the sub-optimal solution [17]. After that, the maximum rate of the worst-user is recalculated according to formula (2). To make the description of the SDR-based iterative D.C. algorithm more readable, the detail is relegated to Algorithm 1.

Convergence: We show the proposed SDR-based iterative D.C. algorithm eventually converges after finite iterations in Appendix.

Complexity: If the number of variables is N, the computational complexity using off-line convex optimization tool, such as conventional interior point method, is $\mathcal{O}(N^{3.5})$[25]. In SDR-based iterative D.C. algorithm, the total number of variables is $(M+NK)^2S + L + 1$. The main computational complexity is $\mathcal{O}\left(\left(\phi_3\left((M+NK)^2S + L + 1\right)\right)^{3.5}\right)$, where ϕ_3 is the average iteration number within error threshold.

Optimality: We have proved the convergence of the SDR-based iterative D.C. algorithm. It can indicate at least a local solution.

Algorithm 1 SDR-based iterative D.C. algorithm

1: **Input:** Set variables initial value $t^{(1)}$, $\mu^{(1)}$ for $\forall l \in \mathcal{L}_s$, $\forall s \in \mathcal{S}$, set iteration index $i = 1$ and maximum tolerance ξ.
2: **while** not convergence **do**
3: Solve (16) by employing CVX in MATLAB.
4: Update $t^{(i+1)} = t^{(*)}$, $\mu_l^{(i+1)} = \mu_l^{(1)}$, and $\mathbf{W}_s^{(i+1)} = \mathbf{W}_s^{(*)}$.
5: Update iteration index $i = i + 1$.
6: Repeat 3–5 until convergence.
7: Output $\mathbf{W}_s^{(*)}$.
8: **end while**
9: **Output:** $\mathbf{w}_{k,s}^{(*)}$ by Gaussian randomization.
10: Recalculate the maximal SINR of worst-user according to formula (2).
11: The achievable rate is calculated by $\log(1 + \mathrm{SINR})$.

3.2. Two-tier convex quadratic-based alternating algorithm

Remark 1. Designing better algorithms for non-convex problem is a key challenge to improve network performance. To combat with non-convexity, most existing works adopt a convex programming framework, such as algorithms based on SDR and SCA [15,17–19,24]. However, the drawbacks of using SDR are twofold. First, this method is computationally expensive especially when the number of transmit antennas is large. Second, the equivalence between the original problem and the recast problem cannot be guaranteed for multiple input single output (MISO) systems because the rank-one constraint has been relaxed. Although the probability of rank-one solution using the SDR method is higher when the number of users/antennas is small, the probability decreases with the increase of users/antennas [27]. In such a situation, the maximum eigenvalue does not dominate the trace of beamformers, resulting in a waste of power.

Therefore, in order to reduce computational complexity and more fully explore the solution space, we propose a TCQA algorithm by decoupling the multi-ratio fractional constraints in this subsection. In the TCQA algorithm, the optimum of the outer optimization problem is derived in closed-form expressions while the inner one is solved by employing iterative D.C. programming. Furthermore, we prove the convergence of the TCQA algorithm and analyze the computational complexity.

To further expose the convexity, we mainly employ the quadratic transform to tackle the multiple-ratio fractional SINR [28,29]. Driven by the theorem 2[28], we develop the following theorem.

Theorem 1. *Given a nonempty constraint set $\mathcal{W}$, numerator function $\left|\mathbf{h}_l^H\mathbf{w}_s\right|^2 \ge 0$, and denominator function $\sum_{j=1,j\neq s}^{S}\left|\mathbf{h}_l^H\mathbf{w}_j\right|^2 + \sigma_l^2 > 0$ for $\forall l \in \mathcal{L}_s$ and $\forall s \in \mathcal{S}$, the max–min-ratio problem*

$$\max_{\mathbf{w}_{k,s}\in\mathcal{W}}\min_{s\in\mathcal{S}}\min_{l\in\mathcal{L}_s}\frac{\left|\mathbf{h}_l^H\mathbf{w}_s\right|^2}{\sum_{j=1,j\neq s}^{S}\left|\mathbf{h}_l^H\mathbf{w}_j\right|^2 + \sigma_l^2} \tag{17}$$

is equivalent to

$$\max_{\mathbf{w}_{k,s}\in\mathcal{W},\mathbf{y},t} \quad t \tag{18a}$$

subject to

$$\begin{aligned}2\mathrm{Re}\left(\mathbf{y}_{l,s}^H\mathbf{h}_l^H\mathbf{w}_s\right) - \mathbf{y}_{l,s}^H\Bigg(&\sum_{j=1,j\neq s}^{S}\mathbf{h}_l^H\mathbf{w}_j\mathbf{w}_j^H\mathbf{h}_l\\ &+\sigma_l^2\Bigg)\mathbf{y}_{l,s} \ge t, \forall l \in \mathcal{L}_s, \forall s \in \mathcal{S},\end{aligned} \tag{18b}$$

where $\mathbf{y}$ refers to a collection of variables $\{\mathbf{y}_{1,1}, \ldots \mathbf{y}_{l_S,S}\}$. t and $\mathbf{y}_{l,s}$ are auxiliary variables introduced for each ratio.

Proof. To make the optimization problem more tractable, we first remove the two minimum operators by introducing the auxiliary variable t and have

$$\frac{\left|\mathbf{h}_l^H \mathbf{w}_s\right|^2}{\sum_{j=1,j\neq s}^{S}\left|\mathbf{h}_l^H \mathbf{w}_j\right|^2+\sigma_l^2} \geq t, \quad \forall l \in \mathcal{L}_s, \forall s \in \mathcal{S}, \tag{19}$$

and then the objective function is replaced with maximizing t. However, the fractional expression is not concave, which cannot define a convex set. We apply the quadratic transform proposed in [28] to deal with the non-convex constraint. The equivalence between (18b) and (19) has been proved in [28]. Therefore, we obtain the result of a theorem. ■

However, coupling between $\mathbf{w}_s$ and $\mathbf{y}_{l,s}$ incurs the non-convexity. In this paper, we employ convex programming to optimize auxiliary variables $\mathbf{y}$ and remaining variables alternately. First, when the beamformers are held fixed, the optimal $\mathbf{y}$ in closed-form expression can be calculated as

$$\mathbf{y}_{l,s}^{*}=\left(\sum_{j=1,j\neq s}^{S}\mathbf{h}_l^H\mathbf{w}_j\mathbf{w}_j^H\mathbf{h}_l+\sigma_l^2\right)^{-1}\mathbf{h}_l^H\mathbf{w}_s \tag{20}$$

for $\forall l \in \mathcal{L}_s$ and $\forall s \in \mathcal{S}$, which can be obtained by solving the first derivative equal to 0.

We then focus on solving the following problem with fixed $\mathbf{y}$

$$\max_{\mathbf{w}_{k,s},q_{k,s},t} \quad t \tag{21a}$$

subject to

$$\text{(4b)–(4e), (18b).} \tag{21b}$$

We can found that (18b) can ensure the convex set because the left side of the formula is a concave function. Adopting a similar approach in handling problem (5), we also introduce the binary selection matrices $\mathbf{J}_k$, so $\left\|\mathbf{w}_{k,s}\right\|^2$ can be rewritten as

$$\left\|\mathbf{w}_{k,s}\right\|^2=\left\|\mathbf{w}_s\mathbf{J}_k\right\|^2. \tag{22}$$

Thus, (4d) can further be expressed as

$$q_{k,s}=\left\|\left\|\mathbf{w}_s\mathbf{J}_k\right\|_2^2\right\|_0. \tag{23}$$

Similar to (12), $\left\|\left\|\mathbf{w}_s\mathbf{J}_k\right\|_2^2\right\|_0=\theta_{s,k}\left\|\mathbf{w}_s\mathbf{J}_k\right\|^2$, where $\theta_{s,k}$ is a constant weight, which is updated iteratively as follows:

$$\theta_{s,k}=\frac{1}{\left\|\mathbf{w}_s^{(i)}\mathbf{J}_k\right\|^2+\delta}. \tag{24}$$

Similarly, if $\mathrm{Tr}\left(\mathbf{W}_s\mathbf{J}_k\right)$ in (15) is changed to $\left\|\mathbf{w}_s\mathbf{J}_k\right\|^2$, (4c) can be expressed as

$$\sum_{s=1}^{S}\left(1-c_{k,f_s}\right)\theta_{s,k}\left\|\mathbf{w}_s\mathbf{J}_k\right\|^2 \leq C_{th}\left(\frac{1}{\log(1+t^{(i)})} -\frac{1}{\log^2(1+t^{(i)})(1+t^{(i)})\ln 2}(t-t^{(i)})\right). \tag{25}$$

Adopting (22), (4b) can be recast as

$$\sum_{s=1}^{S}\left\|\mathbf{w}_s\mathbf{J}_k\right\|^2 \leq P_k, \quad \forall k \in \mathcal{K}. \tag{26}$$

So far, all constraints have been transformed to convex sets. As a result, problem (21) can be recast as

$$\max_{\mathbf{w}_s,t} \quad t \tag{27a}$$

subject to

$$\text{(18b), (26), (25),} \tag{27b}$$

Algorithm 2 Two-tier convex quadratic-based alternating (TCQA) algorithm

1: **Input:** Set initial beamforming vectors and variable value $\mathbf{w}_s^{(1)}$ and $t^{(1)}$, for $\forall s \in \mathcal{S}$, set the out-tier iteration index $j=1$ and the maximum tolerance ξ and τ.
2: **while** not convergence **do**
3: Update the optimal $\mathbf{y}_{l,s}^{(j)}$ according to (20) and set the inner-tier iteration index $i=1$.
4: **while** not convergence **do**
5: Solve (27) by employing CVX.
6: Update $\mathbf{w}_s^{(i+1)}=\mathbf{w}_s^{(*)}$, $t^{(i+1)}=t^{(*)}$.
7: Update iteration index $i=i+1$.
8: Repeat 5–7 until convergence.
9: Output $\mathbf{w}_s^{(*)}$, $t^{(*)}$.
10: **end while**
11: Update iteration index $j=j+1$.
12: Repeat 3–11 until convergence.
13: **end while**
14: **Output:** $\log\left(1+t^{(*)}\right)$.

It is worth noting that problem (27) is already convex and can be solved directly by CVX. The proposed TCQA algorithm for maximizing the worst-user's rate is summarized in Algorithm 2.

Convergence: The TCQA algorithm can converge to a local optimum after finite alternating iterations.

Proof. The proof is divided into two parts. The first part is to prove the convergence of the inner-tier D.C. algorithm, and the other is to prove that the outer-tier alternating algorithm can converge to a stationary point. In Appendix, if constant $\mathbf{y}$ is changed to constant $\mathbf{y}^{(j)}$, we can get $R\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i-1)}\right) \leq R\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i)}\right)$, where $R\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i)}\right)$ denotes the achievable rate at the initial point $\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i-1)}\right)$. This conclusion has been proved in Appendix, so we omit it for the limited space.

As for the outer-tier alternating algorithm, according to (20), $\mathbf{y}$ is derived by optimizing the achievable rate under fixed $\mathbf{Q}$, so we can obtain that

$$R\left(\mathbf{y}^{(j+1)},\mathbf{Q}^{(i)}\right) \geq R\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i)}\right) \tag{28}$$

In the first part, we have proved that $R\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i)}\right) \geq R\left(\mathbf{y}^{(j)},\mathbf{Q}^{(i-1)}\right)$. Summarizing the above statements, we can deduce that the TCQA algorithm converges after finite alternating iterations. ■

Complexity: Adopting the same simulation tool, the complexity of conventional interior point method is $\mathcal{O}(N^{3.5})$ [25]. In problem (27), the number of optimization variables is $(M+NK)S+1$. Therefore, the main computational complexity is $\mathcal{O}\left(\left(\sum_{i=1}^{\phi_1}\phi_2(i)\left((M+NK)S+1\right)\right)^{3.5}\right)$, where ϕ_1 denotes average iteration number when the outer-tier loop stops, and $\phi_2(i)$ is the iteration number when the inner-tier D.C. algorithm converges at the ith outer-tier loop. Compared to the SDR-based iterative D.C. algorithm, its complexity has been reduced significantly.

Optimality: The convergence can indicate at least one local solution. However, simulation results show that our proposed TCQA algorithm outperforms the SDR-based iterative D.C. algorithm in terms of the achievable rate.

4. Simulation results

In this section, we investigate the performance of our two proposed iterative algorithms. Throughout the simulation results, we expect to acquire a comprehensive perspective of the efficacy of the proposed TCQA algorithm. We consider a heterogeneous C-RAN model with a radius of 1000 m in which an MBS is located in the center of the circle and 6 SBSs are evenly distributed around the MBS. The distance

Table 1
Key simulation parameters.

Parameter	Value
Distance-dependent pathloss model	$128.1 + 37.6 \log_{10}(d)$ dB
Transmit antenna power gain at each BS	10 dBi
log-normal shadowing	8 dB
Rayleigh small scale fading	0 dB
Channel bandwidth	10 MHz
Background noise	−172 dBm/Hz
Total backhaul link capacity	100 Mbps
Transmit power of MBS/SBS	8/4 W
Equipped antennas of MBS/SBS	4/2
Number of small BS	6
Normalized content size	10 [30]
Number of users	20
The coverage area of SBS	500 m
MBS caching size	10

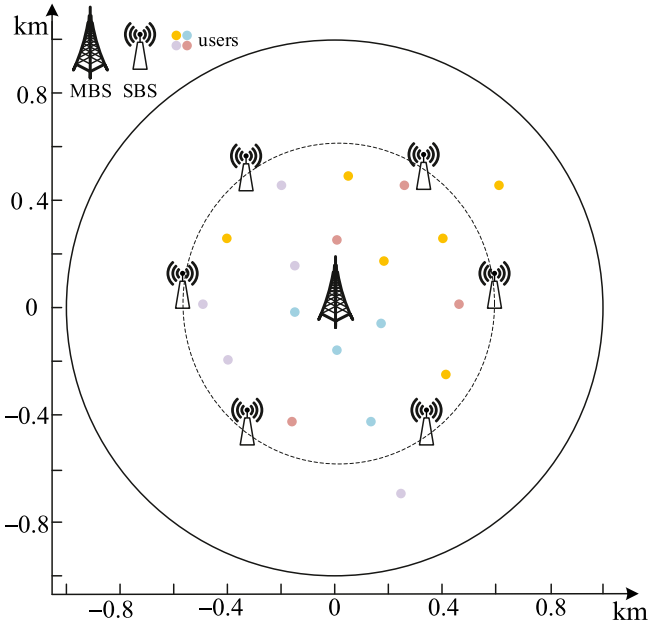


Fig. 2. Simulation scenario is with a MBS and 6 SBSs. Solid circle with the same color denotes the user requesting the same content.

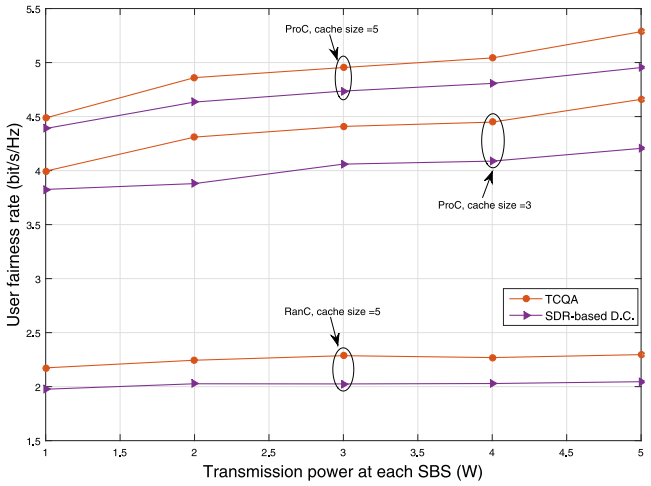


Fig. 3. Maximized minimum rate among all multicast groups versus the transmission power per SBS with different algorithms, caching strategies and cache sizes.

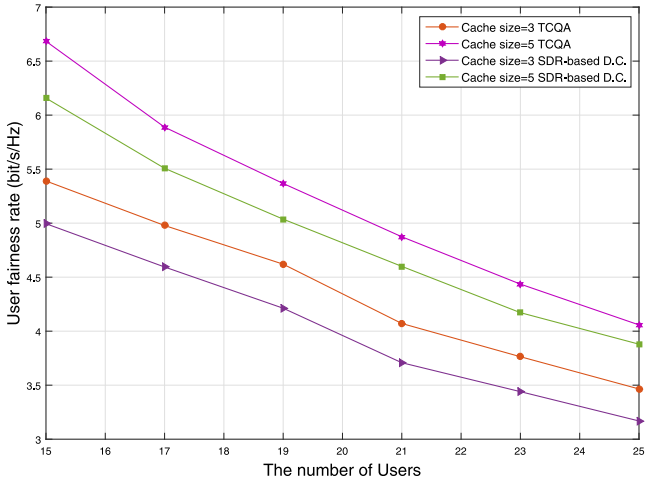


Fig. 4. User fairness rate versus the number of users under different cache sizes and algorithms.

between MBS and SBS is 500 m, as illustrated in Fig. 2. The mobile users are uniformly and independently distributed in the network. Other key simulation parameters are listed in Table 1 unless otherwise specified. Each simulation result is obtained by averaging over 500 Rayleigh-dependent channel realizations. In each simulation, we only randomly generate user location, channel vectors and user requests and content cached by SBS for simulation simplicity.

In addition, we consider the following two different heuristic caching strategies.

- *popularity-aware caching (PopC):* The content requested by users is dominated by a few popularity files. Each SBS caches the most popularity content until its storage space peaks. In such case, the cached content in all SBSs is the same. We assume that content requirements follow Zipf's distribution [31], which can be expressed as

$$p(f) = \frac{f^{-\alpha}}{\sum_{i=1}^{F} i^{-\alpha}}, \quad \forall f \in \{1, \ldots, F\}, \tag{29}$$

  where α is a skewness parameter. The bigger α means higher probability of the most popular files. The content preference distribution approximates uniform distribution when $\alpha \to 0$, which results in the huge backhaul burden. In this paper, $\alpha = 1.5$ [24].

- *Random caching (RanC):* Each SBS caches contents with equal probabilities without considering their popularity distribution.

In Fig. 3, we present the maximized minimum rate among all multicast groups versus the transmission power per SBS in terms of different algorithms, caching strategies and cache sizes. It shows that the proposed TCQA algorithm significantly outperforms the SDR-based iterative D.C. algorithm with both caching strategies, which confirms the effectiveness of our proposed TCQA algorithm. Although the convergence of the TCQA algorithm is strictly slower than superlinear, slower convergence rate can sometimes allow the algorithm to more fully explore the solution space. Interference suffered from other multicast groups increases in the same proportion with the increase of transmission power if the direction of beamformers is unchanged. Thus, the max–min rate rises slowly and eventually reaches saturation. Furthermore, the performance of the random caching strategy is smaller than the popularity caching. The maximized minimum rate remains unchanging because a large number of multicast groups may be formed if users subscribe to the contents with equal probability. Hence, each multicast group will suffer from more interference from other multicast groups, leading to a lower fairness rate.

In Fig. 4, we evaluate the user fairness rate behavior versus the number of users with different cache sizes. The simulation result shows

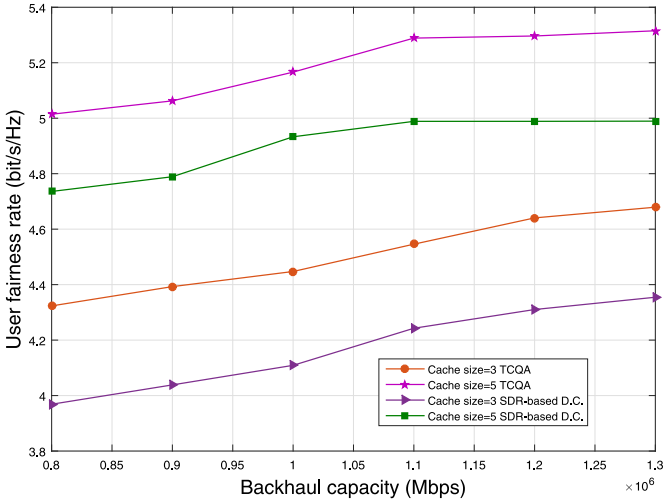


Fig. 5. The maximum of the worst-user rate as a function of backhaul capacity with different cache sizes and algorithms.

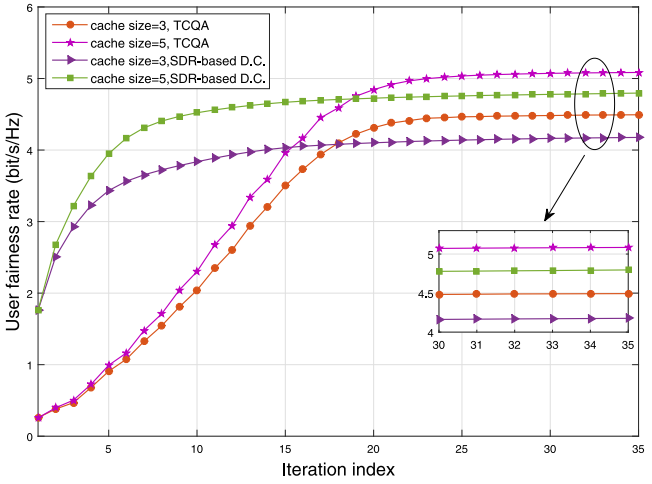


Fig. 6. Convergence performance of our proposed iterative algorithms with different cache sizes.

that the maximum of the worst-user rate is reducing with the increase in the number of users. There are two reasons for this trend. Firstly, the number of multicast groups is small when the number of users is small. Each multicast group suffers less inter-group interference. Secondly, if users requesting different content have very similar channel vectors, no matter how precoding is designed, the signal will suffer considerable interference. Besides, the probability of channel vector similarity increases with the increase of users. Furthermore, SBS caching space size has a positive effect on network performance, which can be explained by that the backhaul link capacity is limited. It also can be seen that the behavior gaps between the TCQA algorithm and the SDR-based iterative D.C. algorithm are gradually diminishing. However, the TCQA algorithm is always better than that of the SDR-based D.C. algorithm.

In Fig. 5, we simulate the user fairness rate versus the backhaul capacity for different cache sizes and algorithms. In this result, we observe that the maximum of the worst-user's rate is increment with the increase of backhaul capacity at the beginning, and then the user fairness rate tends to be stable when cache size = 5. The reason is that the constraint of backhaul links has been released. Specifically, all BS can serve users within its coverage when the backhaul capacity is large enough. An interesting observation, namely the curves with cache size = 3 increases monotonously and approximates that of cache size = 5, can be explained by the fact that more SBS can be included in the BS clustering through backhaul links with the increase of backhaul capacity. When the backhaul capacity is large enough, the performance of the cache size = 3 is close to that of the cache size = 5. That means that the caching capability of SBS will no longer affect system performance.

In Fig. 6, we show the convergence behavior of our two proposed iterative algorithms with different cache sizes. It can be seen clearly that both the SDR-based iterative D.C. algorithm and the TCQA algorithm converge to a steady value after finite iterations, which validates the convergence analysis in the previous section. Specifically, the SDR-based iterative D.C. algorithm eventually converges after around 20 iterations while the TCQA algorithm converges about 25 counts. However, the slower convergence rate has the advantage of fully exploring the solution space. Meanwhile, the computational complexity of the TCQA algorithm is much lower than that of the SDR-based iterative D.C. algorithm.

5. Conclusion

In this work, by jointly designing dynamic BS clustering and beamformers, we aim to maximize the minimum rate for downlink multicast cache-enabled C-RAN. With the per BS power and backhaul cost constraint, the user fairness rate maximization is formulated. To combat the non-convexity, SDP and SCA techniques are employed to handle beamforming coupling while re-weighted l_1-norm approximation is used to solve the discrete BS selection. Then, the SDR-based iterative D.C. algorithm is proposed to cope with the recast problem. For releasing the computational complexity, the TCQA algorithm is devised. In the TCQA algorithm, the inner-tier optimization problem is solved by the D.C. algorithm, and the outer-tier is handled in closed-form expressions. Because practical C-RAN needs to serve a large number of users, algorithms' low complexity is greatly desired. Therefore, our proposed TCQA algorithm with low complexity is easier to implement in large wireless networks. The complexity and convergence of our two proposed algorithms are further analyzed. Simulation results show that the quadratic transform-based alternating algorithm greatly outperforms the SDR-based iterative D.C. algorithm in terms of computational complexity and the maximized minimum rate.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiasi Zhou: Conceptualization, Methodology, Writing - original draft. **Yanjing Sun:** Supervision, Project administration. **Song Li:** Writing - review & editing. **Bin Wang:** Software. **Zhijian Tian:** Writing - review & editing.

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Appendix. Convergence of SDR-based iterative D.C. algorithm

The existing efforts on the convergence of D.C. algorithm is mainly used to that the objective function is D.C. [32,33], which is not suitable for D.C. constraint. In this paper, we prove the convergence of algorithm mainly from the perspective of definition domain. In addition to constraints (7b) and (14), the remaining constraints can define a non-empty convex set. We employ $\mathbf{Q}$ to contain beamformers $\mathbf{W}_s$ and introduced variables t and μ. We then use $f(\mathbf{y},\mathbf{Q})$ and $g(\mathbf{y},\mathbf{Q})$ to model the left and right sides of (7b) and (14), where $\mathbf{y}$ denotes remaining constants. Note that both functions are convex. Therefore, constraints (7b) and (14) can both be equally written as $f(\mathbf{y},\mathbf{Q}) - g(\mathbf{y},\mathbf{Q}) \le 0$. Let $\mathbf{Q}^{(i-1)}$ and $\tilde{g}(\mathbf{y},\mathbf{Q})\big|_{\mathbf{Q}^{(i-1)}}$ denote the ith iteration initial point of the SDR-based iterative D.C. algorithm and first-order Taylor expansion of $g(\mathbf{y},\mathbf{Q})$ at $\mathbf{Q}=\mathbf{Q}^{(i-1)}$, respectively. We can obtain that $\tilde{g}(\mathbf{y},\mathbf{Q})\big|_{\mathbf{Q}^{(i-1)}}$ is lower than $g\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$ within the non-central neighborhood of point $\mathbf{Q}^{(i-1)}$ because $g(\mathbf{y},\mathbf{Q})$ is a convex function. Therefore, the current domain is narrowed down. After using the SDR-based iterative D.C. algorithm, the output local optimum is $R\left(\mathbf{y},\mathbf{Q}^{(i)}\right)$, where $R\left(\mathbf{y},\mathbf{Q}^{(i)}\right)$ denotes the achievable rate at the initial point $\mathbf{Q}^{(i-1)}$. Let $\mathbb{D}\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$ denote the domain defined by $\mathbf{y}$ and $\mathbf{Q}^{(i-1)}$. Unfortunately, we cannot ensure the bigger one between $R\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$ and $R\left(\mathbf{y},\mathbf{Q}^{(i)}\right)$ because we cannot confirm that the initial point $\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$ inside of $\mathbb{D}\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$.

Substituting $\mathbf{Q}^{(i-1)}$ into the first-order Taylor expansion of $g(\mathbf{y},\mathbf{Q})$, we can get

$$f\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right) - \tilde{g}\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)\Big|_{\mathbf{Q}^{(i-1)}} \overset{(a)}{=} f\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right) - g\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right) \overset{(b)}{\le} 0. \tag{A.1}$$

where (a) is hold because the original function is equal to its first-order Taylor expansion at the expansion point while (b) is true because $\mathbf{Q}^{(i-1)}$ is the feasible point at the last iteration. That means $\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$ belongs to $\mathbb{D}\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right)$. Therefore, we can deduce that $R\left(\mathbf{y},\mathbf{Q}^{(i-1)}\right) \le R\left(\mathbf{y},\mathbf{Q}^{(i)}\right)$. Furthermore, since the achievable rate cannot increase without limit, the D.C. algorithm can converge to one local optimum. Hence, the proof is completed. ■

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